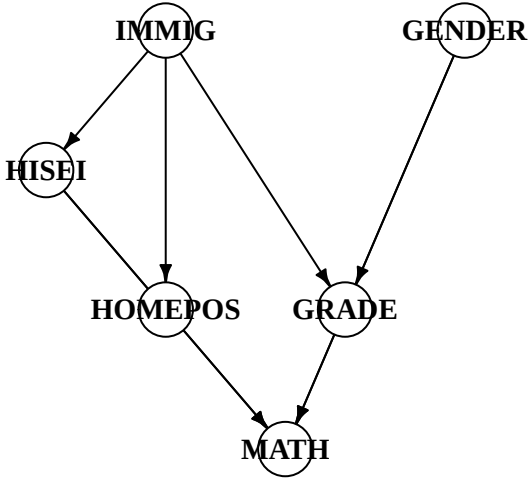PC skeleton + tier ordering